

Supplementary Material for “Excitation-Dependent Identifiability of Latent Higher-Order Structure from Edge Flows”

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This supplement extends the main paper in five directions: **S1** Fano-type converses for *joint* support recovery (including a signal-distribution-agnostic bound and a sub-Gaussian achievability counterpart); **S2** identifiability under *partial edge observation*; **S3** *plug-in* estimation of the variance parameters and a fully adaptive detector; **S4** full proofs of the main paper’s lifted-spark lemma, the three excitation regimes, separation/exponent theorem, and NNLS consistency/failure bound, with exhaustive numerical verification; **S5** construction details of the cyclone vortex-localization study. The released test-suite provides numerical guardrails (not proofs) for the quoted constants; Section S7 maps every figure to its script. Notation follows the main paper: N snapshots, candidate set $\mathcal{T}$ with $p = |\mathcal{T}|$, latent support $\mathcal{S}$ with $|\mathcal{S}| = k$, curl-SNR $\rho = 3\sigma_c^2/\sigma_n^2$, per-coordinate null/alternative variances $v_0 = 3\sigma_n^2$, $v_1 = v_0(1 + \rho)$. Throughout S1–S2 the candidates are pairwise edge-disjoint, so the curl coordinates are independent given $\mathcal{S}$ and the triangle Gram matrix is $\mathbf{G} = 3\mathbf{I}$.

S1 Fano-type converses for joint support recovery

The main paper’s Prop. 2 contains a *single-triangle* converse — the floor $\rho^*(N) = \Theta(\sqrt{\log(1/\delta)/N})$ — together with the Gaussian Fano bound for *joint* recovery of the whole support; Theorem S1.2 below proves the latter, and the signal-agnostic variant extends it.

We first record a reduction used by both proofs.

Lemma S1.1 (Sufficiency of the curl coordinates). *Consider the model of the main paper (Eq. (1)) with isotropic Gaussian noise $\mathbf{n}_t \sim \mathcal{N}(0, \sigma_n^2 \mathbf{I})$, mutually independent $(\mathbf{a}_t, \mathbf{y}_t, \mathbf{h}_t, \mathbf{n}_t)$, and arbitrary laws for $(\mathbf{a}_t, \mathbf{y}_t, \mathbf{h}_t)$ (Gaussianity is needed only for the noise). Then*

$$I(\mathcal{S}; \{\mathbf{f}_t\}_{t=1}^N) = I(\mathcal{S}; \{\mathbf{c}_t\}_{t=1}^N), \quad \mathbf{c}_t = \mathbf{B}_2^\top \mathbf{f}_t.$$

Proof. Decompose $\mathbf{f}_t$ orthogonally into its gradient, curl, and harmonic components. Because the isotropic Gaussian noise has independent projections onto orthogonal subspaces, and the gradient (resp. harmonic, curl) component of $\mathbf{f}_t$ equals $\mathbf{B}_1^\top \mathbf{a}_t$ (resp. $\mathbf{h}_t$, $\mathbf{B}_{2,\mathcal{S}} \mathbf{y}_t$) plus the corresponding independent noise projection, the three components are mutually independent, and only the curl component depends on $\mathcal{S}$. Hence the gradient and harmonic components are independent of $(\mathcal{S}, \text{curl component})$ and carry zero information about $\mathcal{S}$. Finally, on edge-disjoint candidates $\mathbf{B}_2$ has full column rank, so $\mathbf{c}_t$ is a bijective linear image of the curl component. $\square$

Theorem S1.2 (Gaussian Fano converse). *Let $\mathcal{S}$ be uniform on the $\binom{p}{k}$ supports of size k , with Gaussian signals and noise and edge-disjoint candidates. If an estimator achieves $\Pr[\hat{\mathcal{S}} \neq \mathcal{S}] \leq \varepsilon$, then*

$$N \geq \frac{(1 - \varepsilon) \log \binom{p}{k} - \log 2}{k \text{KL}_1(\rho)}, \quad \text{KL}_1(\rho) = \frac{1}{2}(\rho - \log(1 + \rho)) \leq \frac{\rho^2}{4}. \quad (1)$$

In particular $\rho^2 N \gtrsim 4(1 - \varepsilon) \log(p/k)$: joint recovery pays a $\log \binom{p}{k}$ factor where the single-triangle test pays $\log(1/\delta)$.

Proof. By Lemma S1.1 and Fano's inequality, $\Pr[\hat{\mathcal{S}} \neq \mathcal{S}] \geq 1 - (I(\mathcal{S}; \mathbf{C}^N) + \log 2) / \log \binom{p}{k}$ with $\mathbf{C}^N = \{\mathbf{c}_t\}$. Bound the mutual information against the all-inactive reference law P_0 : $I(\mathcal{S}; \mathbf{C}^N) \leq \frac{1}{\binom{p}{k}} \sum_{\mathcal{S}} \text{KL}(P_{\mathcal{S}}^{\otimes N} \| P_0^{\otimes N}) = N \text{KL}(P_{\mathcal{S}} \| P_0)$ for any fixed $\mathcal{S}$ by symmetry. Given $\mathcal{S}$, the coordinates of $\mathbf{c}$ are independent, equal in law to $\mathcal{N}(0, v_1)$ on $\mathcal{S}$ and $\mathcal{N}(0, v_0)$ off $\mathcal{S}$, and the reference has all coordinates $\mathcal{N}(0, v_0)$; hence $\text{KL}(P_{\mathcal{S}} \| P_0) = k \text{KL}(\mathcal{N}(0, v_1) \| \mathcal{N}(0, v_0)) = \frac{k}{2}(\frac{v_1}{v_0} - 1 - \log \frac{v_1}{v_0}) = k \text{KL}_1(\rho)$. Rearranging Fano gives (1); the inequality $\text{KL}_1 \leq \rho^2/4$ is $\log(1 + \rho) \geq \rho - \rho^2/2$. $\square$

Theorem S1.3 (Signal-agnostic Fano converse). *Keep the noise Gaussian, but let the active triangle signals $y_{\tau,t}$ be i.i.d. from an arbitrary zero-mean law with variance σ_c^2 . Then any estimator with error at most ε requires*

$$N \geq \frac{(1 - \varepsilon) \log \binom{p}{k} - \log 2}{\frac{p}{2} \log(1 + \rho k/p)}. \quad (2)$$

Proof. Lemma S1.1 applies verbatim (its proof used only independence and Gaussianity of the noise). With conditionally i.i.d. snapshots, $I(\mathcal{S}; \mathbf{C}^N) \leq N I(\mathcal{S}; \mathbf{c}_1)$, and with conditionally independent coordinates, subadditivity of entropy gives $I(\mathcal{S}; \mathbf{c}) \leq \sum_{\tau} [h(c_{\tau}) - h(c_{\tau} | \mathcal{S})]$. Under the uniform prior each coordinate is active with probability k/p , so $\text{Var}(c_{\tau}) = v_0(1 + \rho k/p)$ and the Gaussian maximum-entropy bound gives $h(c_{\tau}) \leq \frac{1}{2} \log(2\pi e v_0(1 + \rho k/p))$. Conditionally on $\mathcal{S}$, c_{τ} is an independent signal plus $\mathcal{N}(0, v_0)$ noise, so $h(c_{\tau} | \mathcal{S}) \geq \frac{1}{2} \log(2\pi e v_0)$ (adding an independent variable cannot decrease entropy). Summing over the p coordinates yields $I(\mathcal{S}; \mathbf{c}) \leq \frac{p}{2} \log(1 + \rho k/p)$ and Fano concludes. $\square$

Remark S1.4 (How the two converses compare). *Both bounds are valid for Gaussian signals, so one may take their pointwise maximum. For sparse supports ($k \ll p$) and small ρ the Gaussian bound is stronger by a factor $\asymp 1/\rho$ (its denominator is $k \text{KL}_1 \asymp k\rho^2/4$ versus $\asymp k\rho/2$). In the dense regime ($k \asymp p$) the agnostic budget grows only logarithmically in ρ , so it overtakes the Gaussian-KL bound at high SNR (Fig. 1B). Figure 1A shows the $\log p$ effect: at the scale of the $p = 8$ planted-complex setting the joint-recovery floor sits well below the per-triangle Chernoff floor, while at $p = 10^4$ it narrows the gap to a $\approx 2\times$ factor (the curves are not directly comparable in level: Fano is normalized at error $1/2$, the Chernoff floor at $\delta = 0.05$).*

Proposition S1.5 (Sub-Gaussian achievability). *Suppose the curl-coordinate noise and signals are sub-Gaussian with ψ_2 -norms bounded by K times their standard deviations. Then the energy detector with threshold $\gamma = v_0(1 + \rho/2)$ recovers $\mathcal{S}$ exactly with probability $\geq 1 - \delta$ once*

$$N \geq \frac{C K^4}{\min(\rho, \rho^2)/(1 + \rho)^2} \log \frac{2p}{\delta},$$

for a universal constant C : the same $\log p/\rho^2$ scaling as the Gaussian theory, with constants degraded by K^4 .

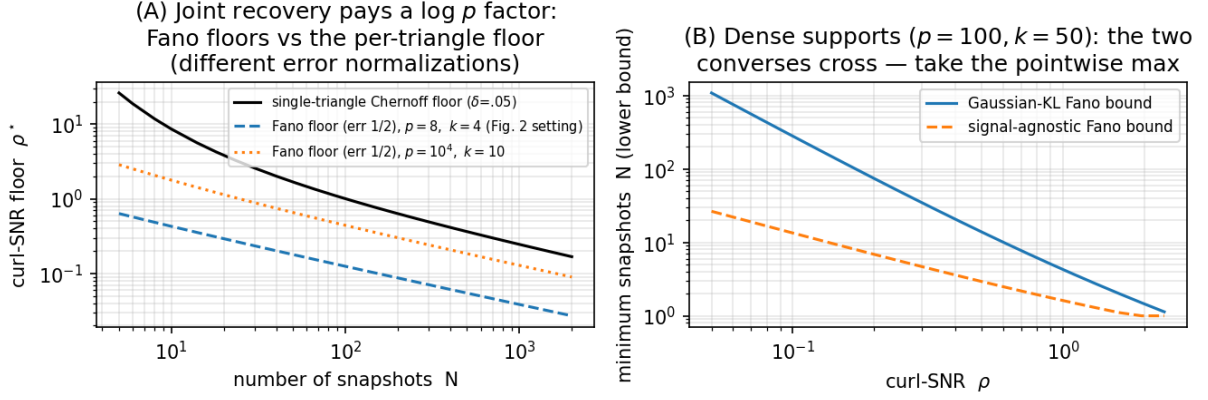


Figure 1: (A) Fano floors (Thm. S1.2, inverted in ρ at error 1/2) against the single-triangle Chernoff floor of the main paper. (B) Dense-support regime: the Gaussian-KL and signal-agnostic bounds cross. Script: `experiments/run_fano.py`.

Proof sketch. Each $c_{\tau,t}^2 - \mathbb{E}c_{\tau,t}^2$ is sub-exponential with ψ_1 -norm $\lesssim K^2 v_i$. Bernstein’s inequality [1, Thm. 2.8.1] gives, for the empirical energy $\hat{E}_\tau = \frac{1}{N} \sum_t c_{\tau,t}^2$,

$$\Pr[|\hat{E}_\tau - v_i| \geq t] \leq 2 \exp\left(-cN \min(t^2/(K^4 v_i^2), t/(K^2 v_i))\right).$$

The two hypotheses are separated by $v_1 - v_0 = v_0 \rho$; taking $t = v_0 \rho/2$ and a union bound over the p candidates yields the claim. $\square$

S2 Partial edge observation

Suppose only a subset $\mathcal{E}_{\text{obs}} \subseteq \mathcal{E}$ of edges is observed — each edge independently with probability q (Bernoulli sampling). Write $\mathbf{f}_t^{\text{obs}}$ for the observed sub-vector.

Proposition S2.1 (Observability for curl statistics). *The curl statistic $c_\tau = \mathbf{B}_2[:, \tau]^\top \mathbf{f}$ of candidate τ is computable from $\mathbf{f}^{\text{obs}}$ iff all three edges of τ are observed. Under Bernoulli(q) sampling each triangle is observable with probability q^3 , independently across edge-disjoint candidates.*

An unobservable *active* triangle is an automatic miss; an unobservable *inactive* one is an automatic correct rejection (the detector defaults to “inactive”). Combining with the exact per-triangle marginal laws (main paper, Prop. 2) gives a closed form on edge-disjoint geometries:

Corollary S2.2 (Exact recovery under edge sampling). *With edge-disjoint candidates, the centered whitened detector at budget N achieves*

$$\Pr[\hat{\mathcal{S}} = \mathcal{S}] = [q^3 (1 - P_{\text{miss}})]^k [1 - q^3 P_{\text{fa}}]^{p-k}, \quad (3)$$

where $P_{\text{miss}}, P_{\text{fa}}$ are the χ_{N-1}^2 error probabilities of the fully observed test.

Proof. Independence of survival (Prop. S2.1) and of the whitened scores ($\mathbf{G} = 3\mathbf{I}$). An active triangle is recovered iff it survives (prob. q^3) and is detected ($1 - P_{\text{miss}}$); an inactive one is mislabeled iff it survives and false-alarms ($q^3 P_{\text{fa}}$). $\square$

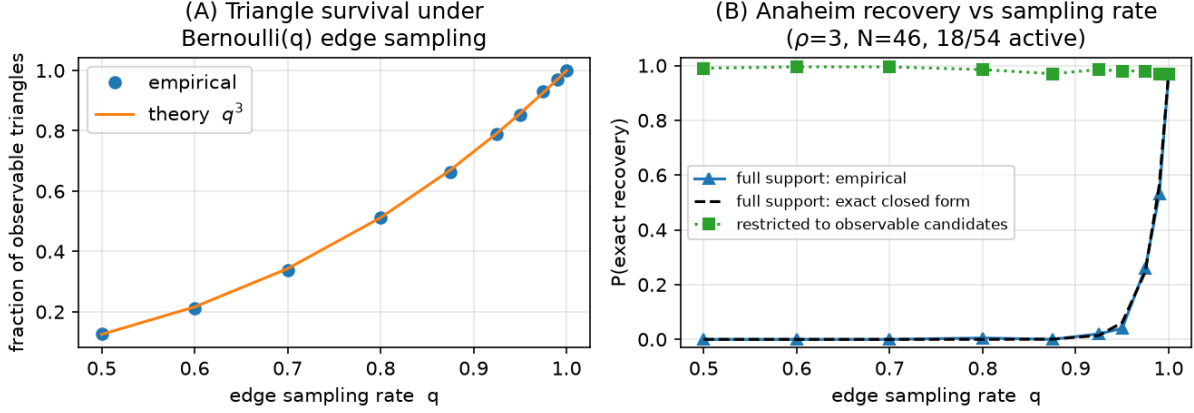


Figure 2: Partial edge observation on Anaheim (real geometry and equilibrium flow background, $\rho = 3$, $N = 46$, 18/54 active). (A) Triangle survival follows q^3 . (B) Full-support recovery collapses along the exact closed form (3); recovery restricted to observable candidates stays ≈ 1 . Script: `experiments/run_partial_sampling.py`.

The q^{3k} factor is brutal: partial sampling multiplies the fully observed recovery probability by $q^{3k} = q^{54}$ at $k = 18$ active triangles — a factor ≈ 0.06 at $q = 0.95$ and ≈ 0.003 at $q = 0.9$ — and the Monte-Carlo cliff in Fig. 2B follows the closed form within sampling error. The restricted curve (recovery on the observable candidates only) stays at ≈ 1 for all q : the bottleneck is purely coverage, not detection.

Remark S2.3 (Scope). *Proposition S2.1 is an operational statement about detectors built on curl statistics — the class this paper analyzes. A triangle with two observed edges still injects signal into those edges’ flows, but that signal is entangled with the unknown gradient nuisance once the exact curl cancellation is unavailable; whether (and at what cost) it can be exploited, e.g. by marginalizing the gradient over the observed sub-graph, is an open question we do not claim to settle. Low-rank flow completion under sampling (cf. graph sampling theory [2]) is the natural attack.*

S3 Plug-in variance estimation and the adaptive detector

The paper’s detectors take (σ_c, σ_n) as known. We now estimate both from the same data. Let $u_\tau = \text{score}_\tau / (\mathbf{G}^{-1})_{\tau\tau}$ denote the normalized whitened scores, so that for inactive τ , $Nu_\tau / \sigma_n^2 \sim \chi_d^2$ ($d = N$ or $N - 1$ if centered), while active scores are stochastically larger.

Estimators.

$$\hat{\sigma}_n^2 = \frac{N \text{med}_\tau(u_\tau)}{\text{med}(\chi_d^2)}, \quad (\text{median, robust to } < 50\% \text{ contamination}) \quad (4)$$

$$\hat{\sigma}_c^2 = \left(\frac{1}{|\hat{\mathcal{A}}|} \sum_{\tau \in \hat{\mathcal{A}}} [\text{score}_\tau - \hat{v}_{0,\tau}] \right)_+, \quad \hat{\mathcal{A}} = \text{Bonferroni screen at level } \alpha, \quad (5)$$

followed by one refinement pass: after a first detection, σ_n is re-fit by the mean of the *detected* off-support scores and σ_c by the on-support excess, and the detector is re-run once. Because the

refit set is selected by the same data, the refit is conditionally biased (missed active triangles inflate $\hat{\sigma}_n$ — measured $\approx +5\%$ at $N = 10$ on the strip benchmark, vanishing for $N \geq 20$); its accuracy is established empirically (Fig. 3A), not by an unbiasedness argument, and Prop. S3.2 applies conditionally on the deterministic error bound ϵ .

Lemma S3.1 (Median consistency envelope). *Assume the candidates are pairwise edge-disjoint, so that the inactive normalized scores are i.i.d. $\sigma_n^2 \chi_d^2/N$ (the DKW step below requires independence; for correlated scores the envelope is validated only empirically). Let a fraction $\pi < 1/2$ of the p candidates be active and let $\epsilon_p = \sqrt{\log(2/\delta)/(2(1-\pi)p)}$ satisfy $\pi + \epsilon_p < 1/2$. Then with probability at least $1 - \delta$,*

$$q_d\left(\frac{1/2-\pi-\epsilon_p}{1-\pi}\right) \leq \frac{N \text{med}_\tau(u_\tau)}{\sigma_n^2} \leq q_d\left(\frac{1/2}{1-\pi} + \frac{\epsilon_p}{1-\pi}\right),$$

where $q_d(\cdot)$ is the χ_d^2 quantile function. Consequently $\hat{\sigma}_n^2/\sigma_n^2 \in [q_d(\frac{1/2-\pi-\epsilon_p}{1-\pi}), q_d(\frac{1/2+\epsilon_p}{1-\pi})]/q_d(1/2)$ — an explicit, numerically computable envelope that shrinks to $\{1\}$ as $p, d \rightarrow \infty$ whenever $\pi < 1/2$.

Proof. Active scores stochastically dominate inactive ones, so the sample median of all p scores is sandwiched between order statistics of the $(1-\pi)p$ inactive scores: at least a $(1/2-\pi)/(1-\pi)$ -fraction and at most a $(1/2)/(1-\pi)$ -fraction of the inactive sample lies below it. The Dvoretzky–Kiefer–Wolfowitz inequality applied to the inactive empirical CDF converts these fractions into population χ_d^2 quantiles up to ϵ_p , with probability $1 - \delta$. $\square$

Remark S3.2 (Plug-in perturbation — empirical observation). *We do not prove a uniform explicit-constant bound on the adaptive detector’s excess error. Heuristically, with plug-in variances $\hat{v}_{i,\tau} = (1 + e_{i,\tau})v_{i,\tau}$, $|e_{i,\tau}| \leq \epsilon < 1/2$, the Bayes threshold $\gamma(v_0, v_1)$ is smooth (main paper, Prop. 2) so a first-order expansion suggests each per-triangle error probability changes by $O(\epsilon)$ and the exact-recovery probability by $O(p\epsilon)$ (union bound); but the constant depends on the χ_d^2 density at the (shifting) threshold and on the $\hat{v}$ concentration, and we have not established a uniform Lipschitz constant over the operating range. We therefore report the effect empirically only. Establishing a non-asymptotic joint selection-plus-estimation bound with explicit constants is left open.*

Empirically (Fig. 3) both variance estimates contract at the $1/\sqrt{N}$ rate and the fully adaptive detector’s recovery curve is indistinguishable from the known-parameter detector for $N \gtrsim 30$ on the confusable strip benchmark; the residual gap at $N \leq 20$ (at most 0.06 after refinement) is the price of estimating a variance from nine medians. These are measurements, not a theorem.

S4 Excitation-dependent identifiability: proofs

This section proves the main paper’s Lemma 1 (lifted spark), Theorem 1 (three excitation regimes), Theorem 2 (separations and exponents), and Theorem 3 (NNLS consistency and failure bound), and records the exhaustive numerical verifications. Here the candidate set is arbitrary (edge-sharing allowed, $\mathbf{B}_2$ possibly rank-deficient); $\rho_2 = \sigma_c^2/\sigma_n^2$ denotes the second-order SNR (the per-triangle curl-SNR is $\rho = 3\rho_2$).

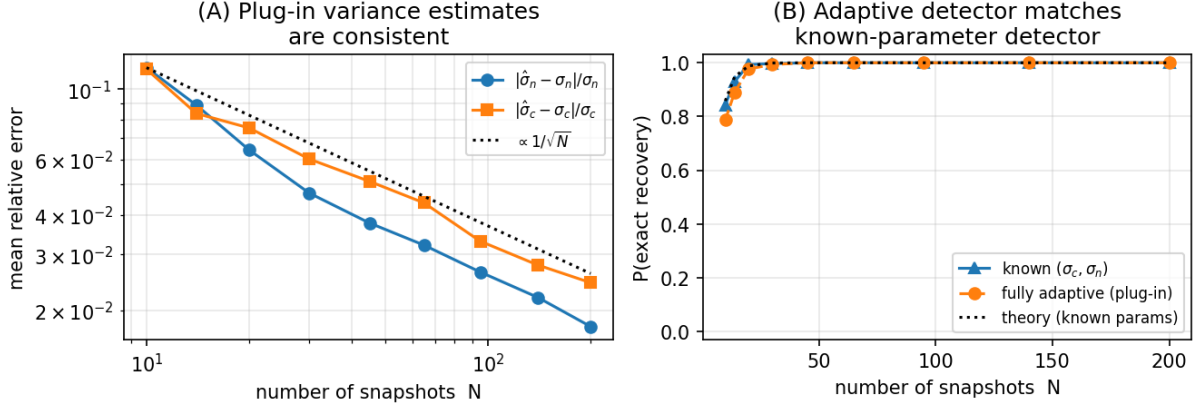


Figure 3: Plug-in estimation on the edge-sharing strip benchmark ($\rho = 8$, $4/9$ active; cf. main Prop. 1). (A) Consistency of (4)–(5) (after refinement). (B) The fully adaptive detector matches the known-parameter detector. Script: `experiments/run_plugin.py`.

S4.1 The lifted spark lemma

Lemma S4.1 (Lifted spark). *Let $\mathcal{T}$ be any set of distinct 3-cliques of a simple graph, with signatures $\mathbf{b}_\tau \in \{0, \pm 1\}^{|\mathcal{E}|}$ (columns of $\mathbf{B}_2$). The matrices $\{\mathbf{b}_\tau \mathbf{b}_\tau^\top\}_{\tau \in \mathcal{T}}$ are linearly independent.*

Proof. Fix $\tau = (i, j, k)$ and its edge pair $e = (i, j)$, $e' = (j, k)$. A pair of distinct edges determines at most one triangle containing both (two edges sharing a vertex fix the third vertex; disjoint edges lie in no common triangle). Hence in $\mathbf{A}(\mathbf{w}) = \sum_\sigma w_\sigma \mathbf{b}_\sigma \mathbf{b}_\sigma^\top$, the (e, e') entry equals $w_\tau \mathbf{b}_\tau(e) \mathbf{b}_\tau(e') = \pm w_\tau$: no other candidate contributes. $\mathbf{A}(\mathbf{w}) = \mathbf{0}$ therefore forces $w_\tau = 0$ for every τ . $\square$

S4.2 The three excitation regimes (main Theorem 1)

With $\mathbf{z} = \mathbf{Q}^\top \mathbf{f}$ ($\mathbf{Q}$ an orthonormal basis of $\text{im } \mathbf{B}_2$, $r = \text{rank } \mathbf{B}_2$, $\mathbf{U} = \mathbf{Q}^\top \mathbf{B}_2$), the gradient term is annihilated unconditionally ($\mathbf{B}_2^\top \mathbf{B}_1^\top = \mathbf{0}$) and the harmonic term by the candidate-orthogonality assumption $\mathbf{h} \in \ker \mathbf{B}_2^\top$, so $\Sigma_z(\mathcal{S}, \Gamma) = \sigma_n^2 \mathbf{I}_r + \mathbf{U}_\mathcal{S} \Gamma \mathbf{U}_\mathcal{S}^\top$; since $\mathbf{b}_\tau \in \text{im } \mathbf{B}_2$, the compression loses nothing ($\mathbf{u}_\sigma^\top \mathbf{u}_\tau = G_{\sigma\tau}$).

Proposition S4.2 ((a) positive diagonal). *If $\Gamma_\mathcal{S} = \text{diag}(\gamma_\tau) > 0$, then $\Sigma_z - \sigma_n^2 \mathbf{I} = \sum_\tau w_\tau \mathbf{u}_\tau \mathbf{u}_\tau^\top$ with $w_\tau = \gamma_\tau \mathbf{1}\{\tau \in \mathcal{S}\}$, and by Lemma S4.1 the coefficients w (hence $\mathcal{S}$ and the weights) are uniquely determined — at any rank deficiency of $\mathbf{B}_2$. On K_n with σ_n unknown, exactly one direction is ambiguous: $\sum_{\tau \in \mathcal{T}} \mathbf{u}_\tau \mathbf{u}_\tau^\top = n \mathbf{I}_r$ (the up-Laplacian of the full complex acts as n on the curl space), so a uniform weight shift trades off against the noise floor; this is verified numerically for K_5 – K_7 .*

Proposition S4.3 ((b) Arbitrary PSD). *For each support the achievable-covariance set is*

$$\mathcal{C}_\mathcal{S} = \{\mathbf{U}_\mathcal{S} \Gamma \mathbf{U}_\mathcal{S}^\top : \Gamma \succeq 0\} = \{\mathbf{M} \succeq 0 : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_\mathcal{S}\}.$$

Consequently, given the population covariance Σ_z (equivalently $\mathbf{M} = \Sigma_z - \sigma_n^2 \mathbf{I}$), the set of supports consistent with it is exactly $\{\mathcal{S} : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_\mathcal{S}\}$. Hence: (i) the realized range $\mathcal{R} = \text{im } \mathbf{M} = \text{im}(\mathbf{U}_\mathcal{S} \Gamma^{1/2}) \subseteq \text{im } \mathbf{U}_\mathcal{S}$ is always identifiable; (ii) the candidate image $\text{im } \mathbf{B}_{2,\mathcal{S}}$ is

not — any support with a larger image and a suitable $\mathbf{\Gamma}$ realizes the same $\mathbf{M}$; (iii) a full-rank $\mathbf{\Gamma} \succ 0$ gives $\mathcal{R} = \text{im } \mathbf{U}_S$ (the converse only when $\mathbf{U}_S$ is injective).

Proof. Set identity: $(\subseteq) \mathbf{U}_S \mathbf{\Gamma} \mathbf{U}_S^\top \succeq 0$ has range in $\text{im } \mathbf{U}_S$; $(\supseteq)$ for PSD $\mathbf{M}$ with $\text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_S$ set $\mathbf{\Gamma} = \mathbf{U}_S^\top \mathbf{M} (\mathbf{U}_S^\top)^\top \succeq 0$, giving $\mathbf{U}_S \mathbf{\Gamma} \mathbf{U}_S^\top = \mathbf{P}_{\text{im } \mathbf{U}_S} \mathbf{M} \mathbf{P}_{\text{im } \mathbf{U}_S} = \mathbf{M}$. The support statement is immediate since $\mathbf{M} \in \mathcal{C}_S \Leftrightarrow \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_S$. Range: $\mathbf{\Gamma} \succ 0 \Rightarrow \mathbf{\Gamma}^{1/2}$ invertible $\Rightarrow \mathcal{R} = \text{im } \mathbf{U}_S$; if $\mathbf{U}_S$ is injective, $\dim \mathcal{R} = \text{rank } \mathbf{\Gamma}$, so equality forces $\mathbf{\Gamma} \succ 0$. $\square$

Remark S4.4 (Strong witness: strictly positive diagonal is not enough). *Positive per-triangle variance does not rescue identifiability; regime (a) needs $\mathbf{\Gamma}$ diagonal. On K_4 with all four faces, $\mathbf{U}_S$ has a one-dimensional kernel spanned by $\mathbf{c} = (1, -1, 1, -1)^\top$ (the tetrahedral relation). Take $\mathbf{v} = \mathbf{e}_1 - \frac{1}{4}\mathbf{c} = (.75, .25, -.25, .25)^\top$ (all entries nonzero) and $\mathbf{\Gamma}' = \mathbf{v}\mathbf{v}^\top$: then $\mathbf{\Gamma}'$ has strictly positive diagonal, rank 1, and $\mathbf{U}_S \mathbf{\Gamma}' \mathbf{U}_S^\top = (\mathbf{U}_S \mathbf{v})(\mathbf{U}_S \mathbf{v})^\top = \mathbf{u}_1 \mathbf{u}_1^\top$ (since $\mathbf{U}_S \mathbf{c} = \mathbf{0}$) — identical to the singleton $\{\tau_1\}$ with unit variance, although $\dim \text{im } \mathbf{B}_{2,S} = 3 \neq 1$. The obstruction is that $\mathbf{\Gamma}'$ is non-diagonal — nonzero off-diagonal entries $v_i v_j \approx \pm 0.19$ (being rank one, $\mathbf{\Gamma}'$ is in fact perfectly correlated) — not any zero variance (the degenerate $\mathbf{\Gamma} = \text{diag}(1, 0)$ example needs an inactive triangle; this one does not). Verified in `tests/test_excitation.py`.*

Proposition S4.5 ((c) projector excitation). *For $\mathbf{\Gamma}_S = \sigma_c^2 (\mathbf{B}_{2,S}^\top \mathbf{B}_{2,S})^+$, $\mathbf{U}_S \mathbf{\Gamma}_S \mathbf{U}_S^\top = \sigma_c^2 \mathbf{P}_{\text{im } \mathbf{U}_S}$ exactly (SVD: $\mathbf{A}(\mathbf{A}^\top \mathbf{A})^+ \mathbf{A}^\top = \mathbf{P}_{\text{im } \mathbf{A}}$). Supports with equal curl image therefore induce identical snapshot distributions — indistinguishable at every SNR and N . Consequently, under a uniform prior over the m supports sharing a given image, no estimator recovers the true one with probability exceeding $1/m$ (the events $\{\hat{\mathbf{S}} = \mathbf{S}_j\}$ are disjoint, so $\Pr[\text{success}] = \frac{1}{m} \sum_j \Pr[\hat{\mathbf{S}} = \mathbf{S}_j] \leq \frac{1}{m}$). For one tetrahedron on K_n the equal-cardinality (size-3) class has $m = 4$ (the four 3-subsets of its faces); the full observational- equivalence class of that image is 5 (adding the 4-face support), which only lowers the ceiling to $1/5$. The main paper's α -sweep recovers a specific size-3 support, so it uses $m = 4$; the estimator's plotted 0 (it returns the wrong cardinality) is itself $\leq 1/m$.*

Remark S4.6 (Deterministic boundary). *Unknown deterministic signals are the rank-one degenerate boundary of regime (b): for equal-image supports the mean families $\{\mathbf{B}_{2,S} \mathbf{y}\} = \{\mathbf{B}_{2,S'} \mathbf{y}'\}$ coincide per snapshot, so no test distinguishes them at any SNR and any N . Tetrahedral confusers realize equal image: the four faces satisfy $\mathbf{b}_{012} - \mathbf{b}_{013} + \mathbf{b}_{023} - \mathbf{b}_{123} = \mathbf{0}$, so any three faces span the same 3-space as any other three; on K_n there are $\binom{n}{4}$ tetrahedra.*

S4.3 The price: exact separations and Chernoff exponents

Lemma S4.7 (Swap separation). *Let $\mathcal{S}' = \mathcal{S} \setminus \{\tau_a\} \cup \{\tau_b\}$ with τ_a, τ_b faces of a common tetrahedron. Then $\Sigma_z(\mathcal{S}) - \Sigma_z(\mathcal{S}') = \sigma_c^2 (\mathbf{u}_a \mathbf{u}_a^\top - \mathbf{u}_b \mathbf{u}_b^\top)$ and*

$$\|\mathbf{u}_a \mathbf{u}_a^\top - \mathbf{u}_b \mathbf{u}_b^\top\|_F^2 = \|\mathbf{u}_a\|^4 + \|\mathbf{u}_b\|^4 - 2(\mathbf{u}_a^\top \mathbf{u}_b)^2 = 9 + 9 - 2 = 16,$$

independent of n , of $|\mathcal{S}|$, and of the hosting tetrahedron (any two faces share exactly one edge, so $\mathbf{u}_a^\top \mathbf{u}_b = G_{ab} = \pm 1$).

Lemma S4.8 (Subset separation — the unrestricted worst case). *Let $\mathcal{S}' = \mathcal{S} \cup \{\tau_4\}$ where τ_4 is the fourth face of a tetrahedron whose other three faces lie in $\mathcal{S}$. Then $\text{im } \mathbf{B}_{2,S} = \text{im } \mathbf{B}_{2,S'}$ (the face signatures are linearly dependent) while $\Sigma_z(\mathcal{S}') - \Sigma_z(\mathcal{S}) = \sigma_c^2 \mathbf{u}_4 \mathbf{u}_4^\top$ with separation $\|\mathbf{u}_4 \mathbf{u}_4^\top\|_F^2 = \|\mathbf{u}_4\|^4 = 9$. Exhaustively over ALL 2^{10} supports of K_5 (`test_exhaustive_k5_confuser_separations`: 45,000+ equal-image distinct pairs), 9 is the unrestricted minimum*

separation, while restricted to pairs of equal cardinality the minimum is 16, attained exactly by the swaps of Lemma S4.7. (An earlier revision of this project wrongly claimed 16 as the unrestricted minimum; the equal-cardinality scoping is essential.)

Proposition S4.9 (Chernoff exponents; rank deficiency is free at the exponent level). *Let C_G be the Chernoff information between $\mathcal{N}(\mathbf{0}, \Sigma_z(\mathcal{S}))$ and $\mathcal{N}(\mathbf{0}, \Sigma_z(\mathcal{S}'))$, and write the DIMENSIONLESS signature Gram $\mathbf{D}_\mathcal{S} = \sum_{\tau \in \mathcal{S}} \mathbf{u}_\tau \mathbf{u}_\tau^\top$ (so $\mathbf{M}_\mathcal{S} = \sigma_c^2 \mathbf{D}_\mathcal{S}$). As $\rho_2 \rightarrow 0$,*

$$C_G = \frac{1}{16} \rho_2^2 \|\Delta \mathbf{D}\|_F^2 (1 + o(1)) = \frac{1}{16\sigma_n^4} \|\Delta \mathbf{M}\|_F^2 (1 + o(1)) = \begin{cases} \rho_2^2(1 + o(1)) & \text{swap,} \\ \frac{9}{16} \rho_2^2(1 + o(1)) & \text{subset,} \end{cases}$$

by Lemmas S4.7–S4.8; the two forms coincide since $\Delta \mathbf{M} = \sigma_c^2 \Delta \mathbf{D}$ (do not write $\rho_2^2 \|\Delta \mathbf{M}\|_F^2$, which double-counts σ_c^4). the optimal error satisfies $P_{\text{err}} \leq e^{-N C_G}$ with matching exponent, and $N^*(\delta) \sim \log(1/\delta)/C_G$. Moreover the subset exponent equals the isolated-triangle detection exponent: $C(\rho) = \rho^2/16$ at $\rho = 3\rho_2$ is $\frac{9}{16}\rho_2^2$ (numerically checked to 3×10^{-5} relative at $\rho_2 = 10^{-5}$, *test_subset_confuser_exponent_equals_isolated_triangle_exponent*). Hence the hardest equal-image decision costs, at the exponent level, exactly one isolated-triangle detection: geometric rank deficiency imposes no exponent penalty, and only the Fano log-multiplicity of the confuser family reflects the geometry.

Proof sketch. Write $\mathbf{E} = \Sigma_0^{-1/2}(\Sigma_1 - \Sigma_0)\Sigma_0^{-1/2}$. The Chernoff exponent between equal-mean Gaussians is $\max_s \frac{1}{2}[\log \det((1-s)\mathbf{W}_0 + s\mathbf{W}_1) - (1-s)\log \det \mathbf{W}_0 - s\log \det \mathbf{W}_1]$; a second-order expansion in $\mathbf{E}$ around $\mathbf{E} = \mathbf{0}$ is maximized at $s = 1/2$ (Bhattacharyya) with value $\frac{1}{16} \text{tr} \mathbf{E}^2 + O(\|\mathbf{E}\|^3)$. Here $\mathbf{E} = \rho_2(\mathbf{u}_a \mathbf{u}_a^\top - \mathbf{u}_b \mathbf{u}_b^\top) + O(\rho_2^2)$ (the shared atoms of $\mathcal{S} \cap \mathcal{S}'$ perturb $\Sigma_0^{-1/2}$ only at $O(\rho_2)$, entering C_G at $O(\rho_2^3)$), so $\text{tr} \mathbf{E}^2 = \rho_2^2 \cdot 16(1 + O(\rho_2))$ by Lemma S4.7. The bound $P_{\text{err}} \leq e^{-N C_G}$ is the Chernoff bound; exponent tightness is classical for i.i.d. observations. $\square$

Proposition S4.10 (Fano over the confuser family). *On K_n , consider the $M = 4\binom{n}{4}$ hypotheses obtained by choosing a tetrahedron and leaving one of its four faces out of an otherwise-active tetrad. Any estimator with error $\leq \epsilon$ over the uniform prior needs*

$$N \geq \frac{(1-\epsilon) \log(4\binom{n}{4}) - \log 2}{\max_{\text{pairs}} \text{KL}(\Sigma_z \| \Sigma'_z)} \asymp \frac{\log n}{\rho_2^2},$$

i.e. full support recovery over the confuser family pays a $\log n$ factor on top of the pairwise budget.

S4.4 Global analytic separations (main Theorem 2)

Theorem S4.11 (Separations on every clique complex). *Let $\mathcal{T}$ be any candidate set of distinct 3-cliques of a simple graph and let $\mathcal{S} \neq \mathcal{S}'$ be binary supports with equal curl image, isotropic excitation $\sigma_c^2 \mathbf{I}$. Then $\|\mathbf{M}_\mathcal{S} - \mathbf{M}_{\mathcal{S}'}\|_F^2 \geq 9\sigma_c^4$, with equality iff $\mathcal{S}, \mathcal{S}'$ differ in exactly one (linearly dependent) face; and $\geq 16\sigma_c^4$ if additionally $|\mathcal{S}| = |\mathcal{S}'|$, with equality iff the difference is a single edge-sharing swap.*

Proof. Write $\mathbf{M}_\mathcal{S} - \mathbf{M}_{\mathcal{S}'} = \sigma_c^2 \sum_\tau c_\tau \mathbf{b}_\tau \mathbf{b}_\tau^\top$ with $c_\tau \in \{-1, 0, +1\}$, $\mathbf{c} \neq \mathbf{0}$. Then

$$\left\| \sum_\tau c_\tau \mathbf{b}_\tau \mathbf{b}_\tau^\top \right\|_F^2 = \sum_{\sigma, \tau} c_\sigma c_\tau G_{\sigma\tau}^2 = \mathbf{c}^\top (9\mathbf{I} + \mathbf{A}) \mathbf{c},$$

since $G_{\tau\tau}^2 = 9$ and $G_{\sigma\tau}^2 \in \{0, 1\}$ equals the share-an-edge adjacency $\mathbf{A}$. Two triangles share an edge iff their vertex triples intersect in two elements, so $\mathbf{A}$ is the adjacency of an induced subgraph of the Johnson graph $J(n, 3)$ (n = number of vertices); the Johnson spectrum $\lambda_j = (3-j)(n-3-j)-j$ gives $\lambda_{\min} \geq -3$, and Cauchy interlacing transfers this to every induced subgraph (for $n < 6$, direct computation gives $\lambda_{\min} \in \{-1, -2\}$). Hence $\text{sep} \geq (9-3)\|\mathbf{c}\|^2 = 6\|\mathbf{c}\|^2$. If $\|\mathbf{c}\|^2 = 1$ (single-face difference) then $\mathbf{c}^\top \mathbf{A} \mathbf{c} = 0$ and $\text{sep} = 9$ exactly; equal image with a single-face difference requires that face's signature to lie in the span of the shared columns (e.g. a hosted tetrahedron's fourth face). If $\|\mathbf{c}\|^2 \geq 2$ then $\text{sep} \geq 12 > 9$, so 9 is the unrestricted minimum. Under $|\mathcal{S}| = |\mathcal{S}'|$, $\sum_\tau c_\tau = 0$ forces $\|\mathbf{c}\|^2$ even and ≥ 2 ; for $\|\mathbf{c}\|^2 = 2$ ($\mathbf{c} = \mathbf{e}_a - \mathbf{e}_b$), $\text{sep} = 18 - 2G_{ab}^2 \geq 16$ with equality iff the two faces share an edge (tetrahedral swap); for $\|\mathbf{c}\|^2 \geq 4$, $\text{sep} \geq 24 > 16$. $\square$

The Chernoff exponents then follow from the Bhattacharyya expansion (Prop. S4.9), for *fixed graph and support pair* as $\rho_2 \rightarrow 0$; in particular the unrestricted worst case (subset confuser, separation 9) has exponent $\frac{9}{16}\rho_2^2(1 + o(1))$, equal to the isolated-triangle detection exponent.

S4.5 NNLS consistency and failure bound (main Theorem 3)

Proof of main Theorem 3. Let $\mathbf{A} = [\text{vec}(\mathbf{u}_\tau \mathbf{u}_\tau^\top)]_\tau$, $\mathbf{s} = \text{vec}(\hat{\Sigma}_z - \sigma_n^2 \mathbf{I})$, true coefficients $\mathbf{w}^* \geq 0$ with $\mathbf{A}\mathbf{w}^* = \text{vec}(\Sigma_z - \sigma_n^2 \mathbf{I})$, and $\mathbf{E} = \text{vec}(\hat{\Sigma}_z - \Sigma_z)$. (i) *Deterministic perturbation.* Optimality of $\hat{\mathbf{w}}$ over the cone $\{\mathbf{w} \geq 0\} \ni \mathbf{w}^*$ gives $\|\mathbf{A}\hat{\mathbf{w}} - \mathbf{s}\|^2 \leq \|\mathbf{A}\mathbf{w}^* - \mathbf{s}\|^2 = \|\mathbf{E}\|^2$; expanding $\mathbf{A}\hat{\mathbf{w}} - \mathbf{s} = \mathbf{A}(\hat{\mathbf{w}} - \mathbf{w}^*) - \mathbf{E}$ yields $\|\mathbf{A}(\hat{\mathbf{w}} - \mathbf{w}^*)\|^2 \leq 2\langle \mathbf{A}(\hat{\mathbf{w}} - \mathbf{w}^*), \mathbf{E} \rangle \leq 2\|\mathbf{A}(\hat{\mathbf{w}} - \mathbf{w}^*)\|\|\mathbf{E}\|$, so $\|\mathbf{A}(\hat{\mathbf{w}} - \mathbf{w}^*)\| \leq 2\|\mathbf{E}\|$ and $\|\hat{\mathbf{w}} - \mathbf{w}^*\|_2 \leq 2\|\mathbf{E}\|/\sigma_{\min}(\mathbf{A})$, with $\sigma_{\min}(\mathbf{A}) > 0$ by Lemma S4.1. (ii) *Exact covariance-error moment.* For i.i.d. $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \Sigma_z)$ and $\hat{\Sigma}_z = \frac{1}{N} \sum_t \mathbf{z}_t \mathbf{z}_t^\top$, Wishart second moments give $\text{Var}(\hat{\Sigma}_{ij}) = (\Sigma_{ii}\Sigma_{jj} + \Sigma_{ij}^2)/N$, hence $\mathbb{E}\|\mathbf{E}\|^2 = \sum_{ij} \text{Var}(\hat{\Sigma}_{ij}) = [(\text{tr } \Sigma_z)^2 + \|\Sigma_z\|_F^2]/N$ exactly. (iii) *Markov.* Thresholding at $w_{\min}/2$ fails only if $\|\hat{\mathbf{w}} - \mathbf{w}^*\|_\infty \geq w_{\min}/2$, so

$$\Pr[\hat{\mathcal{S}} \neq \mathcal{S}] \leq \Pr\left[\|\mathbf{E}\| \geq \frac{\sigma_{\min}(\mathbf{A}) w_{\min}}{4}\right] \leq \frac{16 \mathbb{E}\|\mathbf{E}\|^2}{\sigma_{\min}^2(\mathbf{A}) w_{\min}^2},$$

which is the stated bound. Consistency: $\hat{\Sigma}_z \rightarrow \Sigma_z$ a.s. (LLN), so by (i) $\hat{\mathbf{w}} \rightarrow \mathbf{w}^*$ a.s., and any threshold in $(0, w_{\min})$ recovers $\mathcal{S}$ eventually. Each step is separately Monte-Carlo verified (`test_excitation.py`): the moment identity to 6% relative at $N = 40$, zero violations of (i) in 220 trials (200 at $N=30$, 20 at $N=2000$), and the final bound upper-bounds empirical failure across an (N, ρ_2) grid. $\square$

S4.6 Numerical verification (released tests)

`tests/test_second_order.py` certifies, deterministically or by Monte-Carlo: (i) Lemma S4.1 on K_4 – K_9 , strips, disjoint unions, and random clique complexes; (ii) injectivity of $\mathcal{S} \mapsto \Sigma_z(\mathcal{S})$ *exhaustively* over all 2^4 supports of K_4 ; (iii) the deterministic replication of first-order confusers (residual $< 10^{-9}$); (iv) exhaustively over ALL 2^{10} supports of K_5 (45,000+ equal-image distinct pairs): unrestricted minimum separation = 9 (subset confusers, Lemma S4.8) and equal-cardinality minimum = 16 (swaps, Lemma S4.7); (v) swap C_G/ρ_2^2 within 6×10^{-4} of 1 at $\rho_2 = 10^{-4}$ (and within 5% at $\rho_2 \leq 10^{-2}$), subset C_G matching the isolated-triangle exponent to 3×10^{-5} relative at $\rho_2 = 10^{-5}$, and the Monte-Carlo error of the optimal test within the e^{-NC_G} envelope with empirical exponent decreasing onto C_G ; (vi) exact greedy covariance-atom recovery on rank-deficient K_5 . `tests/test_excitation.py` additionally checks every claim of

the regime and estimator theory: weighted-diagonal identifiability and the K_n unknown-noise ambiguity direction ($\sum_{\tau} \mathbf{u}_{\tau} \mathbf{u}_{\tau}^{\top} = n \mathbf{I}_r$); PSD matching for equal-image supports (Prop. S4.3, residual $< 10^{-9}$, matched $\mathbf{\Gamma}'$ PSD); exact projector-excitation indistinguishability (Prop. S4.5); the separation identity $\mathbf{c}^{\top} (9\mathbf{I} + \mathbf{A}) \mathbf{c}$ and $\lambda_{\min}(\mathbf{A}) \geq -3$ on K_4 – K_7 and random clique complexes; the three steps of main Theorem 3’s proof individually plus the assembled failure bound against empirical failure rates on an (N, ρ_2) grid; the population score-tie of the matched-subspace baseline and its collapse (with NNLS) to chance under projector excitation; and the monotone vanishing of the equal-image gap under the $\mathbf{\Gamma}_{\alpha}$ interpolation, exactly zero at $\alpha = 1$.

S5 The cyclone study: construction details

This section records the methodological details of the main paper’s curl-based vortex-localization study (Fig. 3 there) beyond what fits in the page budget. Positioning: this is *not* filled-triangle topology recovery — the mesh has no equal-image confusers and weather has no latent boolean support; it grounds the sufficient statistic on genuinely unplanted rotational structure.

From winds to edge flows. ERA5 10 m winds (ARCO-ERA5 public archive, 0.703° equian-gular grid, 6-hourly), Western North Pacific (0 – 45°N , 100 – 180°E), Aug–Sep 2020 (244 snapshots; the season contains 13 IBTrACS storms, incl. Bavi, Maysak, Haishen). Mesh nodes subsample every third grid point ($\approx 2.1^{\circ}$); each grid cell splits into two right triangles; edge flow = trapezoidal line integral $\frac{1}{2}(\mathbf{V}_i + \mathbf{V}_j) \cdot (\Delta x, \Delta y)$ in the local equirectangular metric ($\Delta x = R \cos(\bar{\phi}) \Delta \lambda$). The mesh is simply connected, so $\#\text{triangles} = |\mathcal{E}| - |\mathcal{V}| + 1 = \dim \ker \mathbf{B}_1$ and $\mathbf{B}_2$ has full column rank (verified: 1554/1554) — the favorable-geometry regime (a) of the excitation hierarchy, with no equal-image confusers. By Stokes’ theorem the curl statistic of a triangle approximates its circulation, i.e. area-integrated relative vorticity; a sign convention (sorted-vertex traversal vs. geometric orientation) is tracked explicitly in the code and is irrelevant to the energy detector.

Detection. Per 4-day window (16 snapshots), the headline statistic is the temporally centered curl-*energy* score of the main paper’s Prop. 2, area²-normalized to a common mean-vorticity scale; it is parameter-free (no thresholds are set — evaluation is by ROC ranking), and nothing is planted. The decorrelated (GLS, main Prop. 1) score is computed alongside and reported in the released JSON; its $\mathbf{G}^{-1}$ noise amplification hurts pure detection ranking on this near-regular full-rank mesh (internal AUC 0.773 vs. 0.914). Honest statement about motion: cyclones translate 3 – $6^{\circ}/\text{day}$, i.e. ~ 6 – 11 triangle-widths over a 4-day window — the latent support is *not* static. Temporal centering removes the window-mean flow, so what the statistic detects is the *time-varying* circulation that a translating vortex deposits in every triangle it crosses (a perfectly static vortex would be removed by the centering); the i.i.d.-snapshot theory is thus a working model here, mapped through window-pooled ROC evaluation rather than through per-window support recovery. README honesty note 9 states the same.

Validation. Internal: relative vorticity $\zeta = \partial_x v - \partial_y u$ by centered differences on the FULL 0.7° grid (the detector only ever sees the 2.1° mesh flows); per-triangle truth = window-mean $|\zeta|$ averaged over grid points inside the triangle; ROC sweeps the detector score against truth binarized at typhoon-scale $3 \times 10^{-5} \text{ s}^{-1}$. External: IBTrACS v04r01 best-track fixes ($\geq 34 \text{ kt}$ where wind is reported), a triangle labeled positive if a fix falls within 1.5° of its centroid during

the window. The external reference is an external, separately-curated archive of agency best-tracks, not derived from the ERA5 winds our statistic reads — the community’s record of where cyclones actually were. A synthetic Rankine-vortex test (`tests/test_geo.py`) verifies the whole bridge end-to-end: localization at the core, correct cyclonic sign after orientation correction, and rank agreement with the finite-difference vorticity.

Windowing and counts. The season is 244 six-hourly snapshots; the 15 non-overlapping 16-snapshot windows use the first 240 and drop the trailing 4 (a partial window). Every pooled metric is over these 15 complete windows. Denominators, raw vs. eligible: 726 IBTrACS fixes over 13 storms (12 reach ≥ 34 kt) enter labeling; 414 of the $1554 \times 15 = 23,310$ scored triangle-windows are external-positive (prevalence 1.8%), 711 internal-positive.

Robustness and exploratory CIs. The pooled external AUC is stable under the labeling and processing choices: localization radius $1.0\text{--}2.5^\circ$ gives AUC $0.95\text{--}0.87$ (headline $1.5^\circ:0.92$), wind cutoff TD-typhoon ($0\text{--}64$ kt) gives $0.91\text{--}0.92$, window length 2–6 days gives $0.94\text{--}0.90$, and the moving-block CI is insensitive to the block length (1–5 windows: half-widths 0.04). A *storm-cluster* bootstrap that resamples whole storms (rather than time blocks) gives AUC $[0.88, 0.95]$, agreeing with the moving-block interval $[0.879, 0.951]$ — evidence the headline is not an artifact of one resampling assumption. The budget-degradation panel (Fig. 3C) is the mean over *all* 15 windows ($\pm$ window-to-window sd), not a single hand-picked window. *All real-data confidence intervals are exploratory*: one season (WNP 2020), no multi-year or out-of-sample validation; they quantify within-season resampling spread only.

S6 Scaling, GPU Monte-Carlo, and non-oracle selection

The dense (r^2, p) lifted operator of §S4 is fine for $K_4\text{--}K_8$ but has r^2p entries (≈ 2 GB at $p=10^3$, ≈ 1 TB at $p=10^4$). This section reports a *matrix-free* form that scales, a genuine GPU-batched Monte-Carlo, a *non-oracle* threshold with a classical guarantee, and a faithful convex-sparse baseline. Fig. 4 summarizes; all numbers are auto-checked from the `scaling`, `gpu_mc`, and `selection` JSON files in `results/`, and the hardware is recorded in `manifest.json` and `results/hardware/`.

Matrix-free operator. Everything runs through U ($r \times p$): $\mathbf{A}\mathbf{w} = U \text{diag}(\mathbf{w}) U^\top$ and $\mathbf{A}^\top \mathbf{R} = [\mathbf{u}_\tau^\top \mathbf{R} \mathbf{u}_\tau]_\tau$, so the FISTA gradient $\mathbf{A}^\top \mathbf{A}\mathbf{w} - \mathbf{A}^\top \mathbf{C}$ needs no dense operator; the Gram is $\mathbf{A}^\top \mathbf{A} = \mathbf{G} \circ \mathbf{G}$ (Hadamard square of the triangle Gram), giving the Lipschitz constant $\lambda_{\max}(\mathbf{G} \circ \mathbf{G})$ and the bound constant $\sigma_{\min}(\mathbf{A}) = \sqrt{\lambda_{\min}(\mathbf{G} \circ \mathbf{G})}$ matrix-free. Cost is $O(r^2p)$ per iteration, $O(rp + r^2)$ memory; since $r = \text{dof} \cdot p$, cost $\sim \text{dof}^2 p^3$ and memory $\sim \text{dof}^2 p^2$, so strong rank deficiency is both the interesting and the affordable regime. The matrix-free FISTA and an exact Lawson–Hanson active-set solver reproduce the dense `scipy.nnls` support with 0 mismatches across $K_4\text{--}K_8$ (relative weight error $\leq 10^{-10}$). On a 128-core Xeon the estimator recovers at $p=10^3$ in 0.49s/solve (dof 0.5, $N=400$, 100% exact recovery) where the dense operator would be 2.0 GB (Fig. 4A).

GPU-batched Monte-Carlo. Sampling B curl-domain covariances on-device and solving all B NNLS problems with batched einsum GEMMs turns the Monte-Carlo into dense GPU work. On an RTX 4080 (32 GB), the fully-batched $p=10^3$ run ($B=10^3$) gives exact-recovery

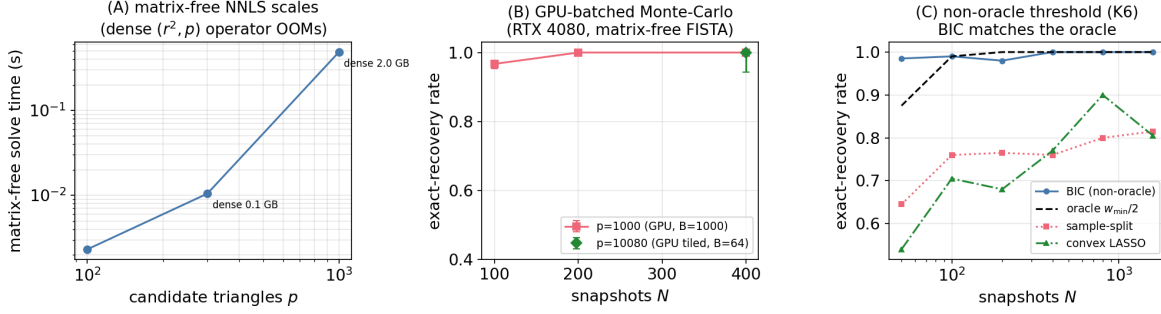


Figure 4: (A) Matrix-free NNLS solve time vs. candidate count p (planted overlapping- K_s complexes, dof 0.5); the dense (r^2, p) operator size is annotated. (B) GPU-batched exact recovery vs. N at $p=10^3$ ($B=10^3$ fully batched) and a tiled $p \approx 10^4$ feasibility point, on an RTX 4080. (C) The non-oracle BIC threshold matches — and at small N beats — the oracle $w_{\min}/2$ rule, and the NNLS estimator beats a convex lifted-LASSO baseline.

0.967/1.00/1.00 at $N=100/200/400$ (≈ 65 s, 6.0 GB peak), and a tiled $p=10^4$ feasibility point ($r=1890$, dof 0.19; $B=64$) recovers 1.00 in 133 s at 3.0 GB (Fig. 4B). The batched GPU FISTA matches the CPU matrix-free path on a shared cross-check. (The $p=10^4$ run is deliberately a small- B feasibility measurement; its wide CI is labeled as such.)

Non-oracle threshold. The main paper’s $w_{\min}/2$ rule uses the true $w_{\min}$ and σ_n (an oracle setting). Replacing both: estimate σ_n from the median eigenvalue of $\hat{\Sigma}_z$, then walk the NNLS path (nest the support by decreasing $\hat{w}$) and select the support minimizing the Gaussian-model BIC

$$\text{BIC}(\mathcal{S}) = N [\log \det \Sigma_{\mathcal{S}} + \text{tr}(\Sigma_{\mathcal{S}}^{-1} \hat{\Sigma}_z)] + |\mathcal{S}| \log N.$$

This inherits the classical BIC selection consistency for Gaussian covariance models (it selects $\mathcal{S}^*$ with probability tending to 1 as $N \rightarrow \infty$ at fixed p) and is confirmed numerically (Fig. 4C): on K_6 the BIC rule matches the oracle at every N and *beats* it at $N=50$ (0.985 vs. 0.875), using no $w_{\min}$; the σ_n estimate is within 2–7%. A sample-splitting variant is provided as an assumption-light fallback.

Convex-sparse baseline. A non-negative lifted LASSO $\min_{w \geq 0} \frac{1}{2} \|C - Aw\|_F^2 + \lambda \|w\|_1$ (soft-threshold FISTA, λ selected by BIC) in the sparse-covariance / sparse-PCA lineage (Wainwright 2009; Amini–Wainwright 2009; Berthet–Rigollet 2013; the main paper’s references) is a faithful convex baseline: on K_6 ($k=3$) it recovers only 0.54–0.90 across N , below both the NNLS/BIC and oracle rules (Fig. 4C) — the assumption-matched, provably consistent NNLS estimator outperforms the generic convex-sparse approach in-regime.

S7 Reproducibility

All figures regenerate on CPU from fixed seeds via `experiments/run_all.py`; individually:

- Fig. 1 $\leftarrow$ `run_fano.py` (pure theory, seconds);
- Fig. 2 $\leftarrow$ `run_partial_sampling.py` (≈ 4 min);

- Fig. 3 $\leftarrow$ `run_plugin.py` (≈ 3 min);
- the main paper’s achievability grid and α -sweep $\leftarrow$ `run_second_order.py` (≈ 15 – 25 min; K_4 – K_8 , random supports, full (N, ρ_2) grid, Wilson intervals, all cells in `results/second_order.json`);
- the main paper’s vortex-localization study $\leftarrow$ `run_real_cyclone.py` (≈ 3 min from the cached ERA5/IBTrACS files).

The test-suite (`pytest -q`, 52 tests) checks: the Fano bounds against the empirical recovery boundary of the phase-transition experiment (`run_phase_transition.py`, `repo/supplement`; a converse must lower bound it), the closed form (3) against Monte-Carlo, the envelope of Lemma S3.1, the adaptive-vs-known recovery gap, every claim of Section S4 (spark, exhaustive K_4 injectivity, separation 16, $C_G/\rho_2^2 \rightarrow 1$, Chernoff envelope, rank-deficient recovery), and the wind-to-edge-flow bridge (mesh full rank by Euler; a synthetic Rankine vortex localizes at the correct triangles with the correct sign and matches the finite-difference vorticity ranking).

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